

1 Definitions

- ◇ Differentiator: ks , where k is a constant
- ◇ Integrator: $\frac{k}{s}$, where k is a constant
- ◇ Dominant poles are the **rightmost** poles.

2 Transfer function

Feedback path transfer function: $H(s)$
Negative or unity ($H(s) = 1$) feedback:

$$CLTF = \frac{\text{Forward path TF}}{1 + H(s)(\text{Forward path TF})}$$

Positive feedback:

$$CLTF = \frac{\text{Forward path TF}}{1 - H(s)(\text{Forward path TF})}$$

3 Mason's gain formula

$$\text{Formula: } \frac{C(s)}{R(s)} = \sum_{k=1}^n \frac{T_k \Delta_k}{\Delta}$$

T_k : Gain of the k^{th} forward path.

Determinant (Δ):

◇ Sum the gain for all individual loops.

◇ Sum the gain products for 2 non-touching loops, then 3, then 4, and so on, **adding** the **even-numbered** gain products and subtracting the **odd-numbered** gain products.

$\Delta = 1 - \text{Sum of the previous 2 steps}$

Associated path factor (Δ_k):

◇ The part of Δ **not touching** the k^{th} forward path.

3.1 Steps

◇ Get the gains for all **forward paths**.

◇ Get the gains for all **loops**.

◇ Get the gains for all **non-touching loops**, then use the formula to solve.

4 Mathematical modelling

Newton's 2nd law: $F = ma = I\alpha$

Torque: $T = Fd_{\perp}$

Spring: $\mathcal{L}\{kx\} = KX(s)$

Damper: $\mathcal{L}\{c\dot{x}\} = \mathcal{L}\{c\dot{x}\} = csX(s)$

Mass acceleration due to gravity:

$$\mathcal{L}\{Ma\} = \mathcal{L}\{M\ddot{x}\} = Ms^2X(s)$$

Moment of inertia:

$$\mathcal{L}\{I\ddot{\theta}\} = \mathcal{L}\{I\ddot{\theta}\} = Is^2\Theta(s)$$

Gear ratio (direction is important):

$$n = \frac{\text{Driving}_1}{\text{Driven}_2} = \frac{r_1}{r_2} = \frac{N_1}{N_2} = \frac{T_1}{T_2} = \frac{\omega_2}{\omega_1} = \frac{\theta_2}{\theta_1}$$

Gear ratio for inertia and damping: n^2

5 Poles and stability

Characteristic equation of $G(s) = \frac{N(s)}{D(s)}$:

Denominator: $D(s) = 0$

◇ Transfer function poles are the **roots of the characteristic equation**.

5.1 BIBO stability

◇ A system is bounded-input bounded-output (BIBO) stable if the output is bounded when the input is bounded.

◇ **Stable systems** have poles with **negative real parts**, like -1 or $-1+j$.

◇ **Partially-stable systems** have **imaginary poles**, like $-j$ or $-2j$.

◇ **Unstable systems** have poles with **positive real parts**, like 1 or $1-2j$.

5.2 Routh-Hurwitz criterion

◇ If there are **no sign changes** in the first column, the system is **stable**.

◇ If one of the elements in the first column is 0, replace the zero with ε , which is a small positive number. The system is **partially stable** if there are **no sign changes**, and **partially unstable** if there are **sign changes** in the first column.

◇ If an entire row of elements are 0, form an equation with the row s above it and **differentiate** the equation with respect to s . The system is **partially stable** if there are **no sign changes**, and **partially unstable** if there are **sign changes** in the first column.

5.2.1 Method

For the characteristic equation:

$$s^4 + 5s^3 + 10s^2 + 50s + 100 = 0$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{5 \times 10 - 1 \times 50}{5} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & 0 & \frac{5 \times 100 - 1 \times 0}{5} & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{50 \times \varepsilon - 5 \times 100}{\varepsilon} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{50 \times \varepsilon - 5 \times 100}{\varepsilon} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{50 \times \varepsilon - 5 \times 100}{\varepsilon} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{50 \times \varepsilon - 5 \times 100}{\varepsilon} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \frac{50 \times \varepsilon - 5 \times 100}{\varepsilon} & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

Since ε is a small positive number, let $\varepsilon = 1$.

$$\begin{array}{r|rrrr} s^4 & 1 & 10 & 100 & \\ s^3 & 5 & 50 & 0 & \\ s^2 & \varepsilon & & & \\ s^1 & \frac{50(1) - 500}{\varepsilon} = -450 & & & \\ s^0 & \frac{50(1) - 500}{\varepsilon} = -450 & & & \end{array}$$

Since there is a sign change in the first column, the system is **partially unstable**.

6 Second order system response

Impulse input: $R(s) = A$

Step input: $R(s) = \frac{A}{s}$

Ramp input: $R(s) = \frac{A}{s^2}$

Sinusoidal input: $R(s) = \frac{\omega}{s^2 + \omega^2}$

General formula:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Undamped or natural frequency: ω_n

Damping ratio (ζ):

$0 < \zeta_{\text{under}} < 1$, $\zeta_{\text{crit}} = 1$, $\zeta_{\text{over}} > 1$

Damped natural frequency:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

Angle:

$$\phi = \arctan \frac{\omega_d}{\zeta\omega_n}$$

Rise time:

$$t_r = \frac{\pi - \phi}{\omega_d}$$

Peak time:

$$t_p = \frac{\pi}{\omega_d}$$

Maximum overshoot:

$$M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

Settling time:

$$t_s = 3T = \frac{3}{\zeta\omega_n} \text{ for 5\% criterion}$$

$$t_s = 4T = \frac{4}{\zeta\omega_n} \text{ for 2\% criterion}$$

7 Steady state analysis

DC gain, or response to unit step input:

$$\lim_{s \rightarrow 0} sG(s) \left(\frac{1}{s} \right) = \lim_{s \rightarrow 0} G(s)$$

Position error constant (k_{pos}):

$$k_{pos} = \lim_{s \rightarrow 0} G(s)$$

$$e(\infty) = \frac{1}{1 + k_{pos}}$$

Velocity error constant (k_{vel}):

$$k_{vel} = \lim_{s \rightarrow 0} sG(s)$$

$$e(\infty) = \frac{1}{k_{vel}}$$

Acceleration error constant (k_{acc}):

$$k_{acc} = \lim_{s \rightarrow 0} s^2 G(s)$$

$$e(\infty) = \frac{1}{k_{acc}}$$

8 Root locus

Open-loop poles: p

Open-loop zeros: z

Characteristic equation:

$$1 + K \frac{N(s)}{D(s)} = 0, \text{ where } K \frac{N(s)}{D(s)} \text{ is the OLTF}$$

Starting points:

Denominator: $D(s) = 0$

Ending points:

Numerator: $N(s) = 0$

Asymptote angle, open-loop zeros at ∞ :

$$\angle s = \frac{\pi(1+2q)}{N_{\text{poles}} - N_{\text{zeros}}}, \text{ where } q = 0, 1, 2, \dots$$

Asymptote location:

$$\sigma_a = \frac{\sum(\text{Re}(p)) - \sum(\text{Re}(z))}{N_{\text{poles}} - N_{\text{zeros}}}$$

Break-in or break-out points:

$$\frac{dK}{ds} = 0, \text{ reject points not on the locus}$$

Phase condition:

$$\sum \angle(s+z) - \sum \angle(s+p) = \pm\pi$$

8.1 Departure and arrival angles

◇ Take a point infinitesimally close to one of the **starting points** or **poles** for the **departure angle**, or one of the **ending points** or **zeros** for the **arrival angle**.

◇ Calculate the angles to the point for each pole and zero and use the phase condition above to find the remaining angle.

◇ The angle for the lines must **start from the right side** of the point, rotating **anti-clockwise**.

8.2 Intersection of the imaginary axis

Substitute $s = j\omega$ into the characteristic equation and solve for K and ω .

8.3 Drawing the root locus

◇ Number all the points on the **real axis**, starting from 1, and from the **right side** going to the **left**. ◇ Every **odd-numbered** point on the **real axis**, regardless of whether it is a zero or a pole, has a root locus to the **left** of it.

9 Controllers

PD controller:

$$G_c(s) = K_P + K_D s = K_D(s + z_c)$$

PI controller:

$$G_c(s) = K_P + \frac{K_I}{s} = \frac{K_P(s + z_c)}{s}$$

◇ When z_c is small ($0 < z_c < 1$), pole-zero cancellation occurs and the transient response is only slightly affected.

Phase (lead or lag) compensator:

$$G_c(s) = K \frac{s+z}{s+p}$$

PID controller:

$$G_c(s) = K_P + K_D s + \frac{K_I}{s}$$

◇ PD to tune transient response.

◇ PI to eliminate steady-state error.

Lead-lag compensator:

$$G_c(s) = K_1 \frac{s+z_1}{s+p_1} + K_2 \frac{s+z_2}{s+p_2}$$

◇ Lead compensator to tune transient response.

◇ Lag compensator to reduce steady-state error.

9.1 Designing controllers

◇ Form equation in s using desired roots.

◇ Use arbitrary roots if there are higher powers of s in the transfer function.

◇ Equate the denominator of the closed-loop transfer function with the controller to the equation with the desired roots and solve for the unknowns.

10 Bode plots

◇ Break down complex functions into **individual functions** and graph the asymptotes individually before adding them to get the net asymptote.

◇ Magnitudes must always be in decibels, i.e. $20 \lg \omega$.

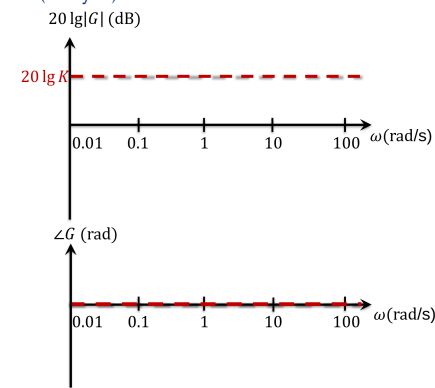
◇ Corner frequencies are always $0.1a$, a , and $10a$.

◇ Must always show peaks if the damping ratio (ζ) is less than $\frac{1}{\sqrt{2}}$.

10.1 Graphs

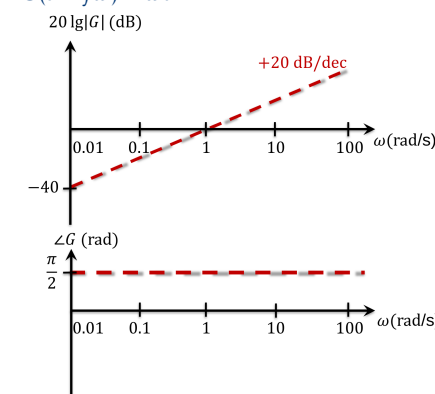
Constants ($G(s) = K$):

$$G(s = j\omega) = K e^{j0}$$



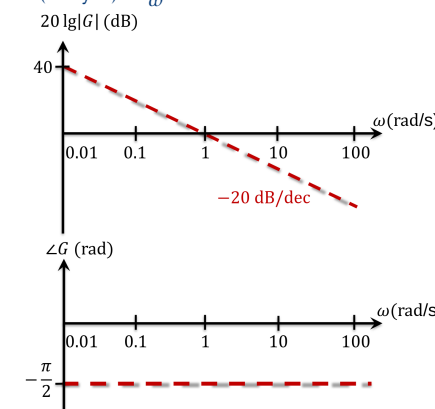
Differentiator ($G(s) = s$):

$$G(s = j\omega) = \omega e^{j\frac{\pi}{2}}$$



Integrator ($G(s) = \frac{1}{s}$):

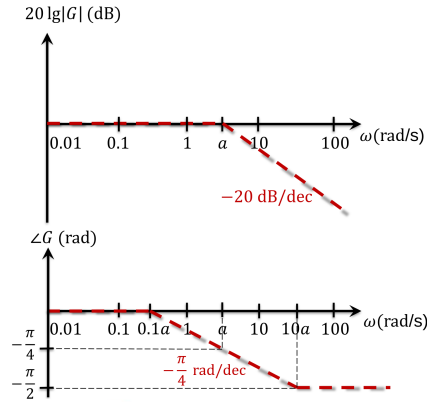
$$G(s = j\omega) = \frac{1}{\omega} e^{j(-\frac{\pi}{2})}$$



First order pole ($G(s) = \frac{a}{s+a}$):

When $\omega \ll a$: $G(s = j\omega) = 1e^{j0}$

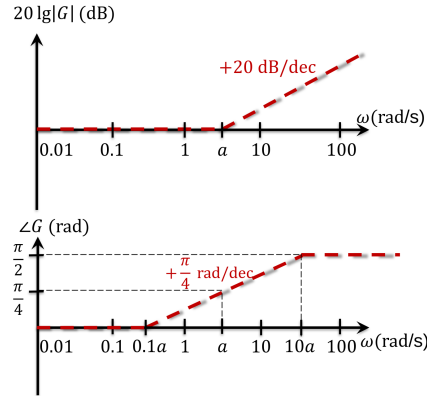
When $\omega \gg a$: $G(s = j\omega) = \frac{a}{\omega} e^{j(-\frac{\pi}{2})}$



First order zero ($G(s) = \frac{s+a}{a}$):

When $\omega \ll a$: $G(s = j\omega) = 1e^{j0}$

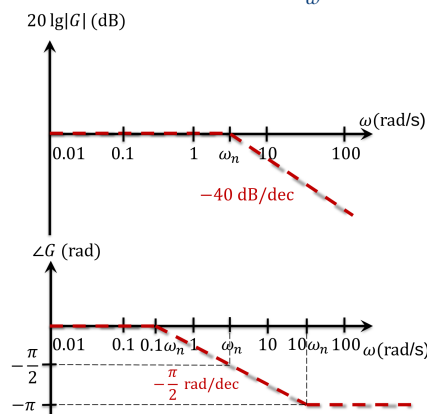
When $\omega \gg a$: $G(s = j\omega) = \frac{\omega}{a} e^{j\frac{\pi}{2}}$



Second order pole ($G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$):

When $\omega \ll \omega_n$: $G(s = j\omega) = 1e^{j0}$

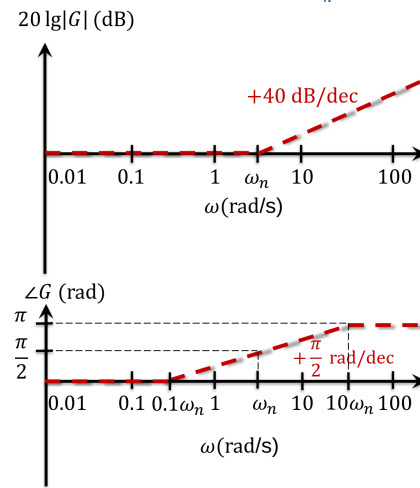
When $\omega \gg \omega_n$: $G(s = j\omega) = \frac{\omega_n^2}{\omega^2} e^{j(-\pi)}$



Second order zero ($G(s) = \frac{s^2 + 2\zeta_n s + \omega_n^2}{\omega_n^2}$):

When $\omega \ll \omega_n$: $G(s = j\omega) = 1e^{j0}$

When $\omega \gg \omega_n$: $G(s = j\omega) = \frac{\omega^2}{\omega_n^2} e^{j\pi}$



11 Bode plot analysis

Low-pass filter:

$$H(s) = \frac{K}{s^2 + bs + a}$$

Band-pass filter:

$$H(s) = \frac{Ks}{s^2 + bs + a}$$

High-pass filter:

$$H(s) = \frac{Ks^2}{s^2 + bs + a}$$

Resonance, $\zeta < \frac{1}{\sqrt{2}}$:

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

Gain margin at $\angle G = \pi$:

$$GM = 0 - |G(s = j\omega_{GM})|_{dB}$$

Phase margin at $|G| = 0 dB$:

$$\gamma = \angle G(s = j\omega_\gamma + \pi)$$

Unity feedback is only stable if:

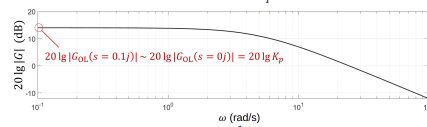
$$\gamma > 0 \text{ and } GM > 0$$

Closed-loop damping ratio (ζ_{cl}):

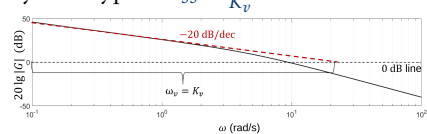
$$\gamma \uparrow \text{ implies } \zeta_{cl} \uparrow$$

11.1 Steady state errors

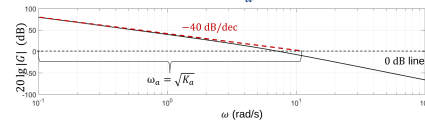
System type 0: $e_{ss} = \frac{1}{1+K_p}$



System type 1: $e_{ss} = \frac{1}{K_v}$



System type 2: $e_{ss} = \frac{1}{K_a}$



11.2 Steady-state output of complex functions

◇ Set the frequency to the ω of the given input function.

◇ Draw a vertical line from the phase graph to the magnitude graph.

◇ Get the phase and magnitude from the graph.

11.3 Transfer function approximation (magnitude plot)

◇ Check, in the low frequency region, for slopes of $20n$ and $-20n$ for n number of differentiators and integrators respectively.

◇ Search for peaks for second order poles and zeros, and find the frequencies where the slopes start to change, those frequencies will be the corner frequencies.

◇ Sharp peaks can be considered the natural frequency, while gentler ones require using $\omega_R = \omega_n \sqrt{1 - 2\zeta^2}$ with the estimated damping ratio to find the natural frequency. Low damping ratio is $\zeta < 0.1$.

◇ Read the magnitude at the lowest frequency (ω_{low}).

◇ If there is no slope, the DC gain is the value, and substitute $s = 0$ into the transfer function to solve for K .

◇ Otherwise, use that value and substitute $s = j\omega_{low}$ instead.

12 Maths

12.1 Approximation

1 + Small Value = 1

1 + Big Value = Big Value

12.2 Laplace transforms

$\mathcal{L}\{k\} = \frac{k}{s}$, where k is a constant

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

$$\mathcal{L}\{at\} = \frac{a}{s^2}$$

$$\mathcal{L}\left\{\frac{at^n}{n!}\right\} = \frac{a}{s^{n+1}}$$

$$\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}$$

$$\mathcal{L}\{te^{-at}\} = \frac{1}{(s+a)^2}$$

$$\mathcal{L}\left\{\frac{1}{(n-1)!} t^{n-1} e^{-at}\right\} = \frac{1}{(s+a)^n}$$

$$\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}\{t \sin \omega t\} = \frac{2\omega s}{(s^2 + \omega^2)^2}$$

$$\mathcal{L}\{t \cos \omega t\} = \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$$

$$\mathcal{L}\{e^{-at} \sin \omega t\} = \frac{\omega}{(s+a)^2 + \omega^2}$$

$$\mathcal{L}\{e^{-at} \cos \omega t\} = \frac{s+a}{(s+a)^2 + \omega^2}$$

$$\mathcal{L}\{1 - e^{-at}\} = \frac{a}{s(s+a)}$$

$$\mathcal{L}\left\{t - \frac{1-e^{-at}}{a}\right\} = \frac{a}{s^2(s+a)}$$

$$\mathcal{L}\{e^{-at} - e^{-bt}\} = \frac{b-a}{(s+a)(s+b)}$$

$$\mathcal{L}\{be^{-bt} - ae^{-at}\} = \frac{(b-a)s}{(s+a)(s+b)}$$

$$\mathcal{L}\{(1-at)e^{-at}\} = \frac{s}{(s+a)^2}$$

$$\mathcal{L}\left\{1 - \frac{b}{b-a}e^{-at} + \frac{a}{b-a}e^{-bt}\right\} = \frac{ab}{s(s+a)(s+b)}$$

$$\mathcal{L}\left\{1 - e^{-at} \left(\cos bt + \frac{a}{b} \sin bt\right)\right\} = \frac{a^2 + b^2}{s[(s+a)^2 + b^2]}$$

$$\mathcal{L}\left\{\frac{e^{-at}}{(b-a)(c-a)} + \frac{e^{-bt}}{(c-a)(a-b)} + \frac{e^{-ct}}{(a-c)(b-c)}\right\} = \frac{1}{(s+a)(s+b)(s+c)}$$

$$\mathcal{L}\left\{\frac{\omega}{\sqrt{1-\zeta^2}} e^{-\zeta\omega t} \sin \omega \sqrt{1-\zeta^2} t\right\} = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2}$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{\frac{\omega}{\sqrt{1-\zeta^2}} e^{-\zeta\omega t} \sin \omega \sqrt{1-\zeta^2} t\right\} = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2}$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\mathcal{L}\left\{1 - \frac{1}{1-\zeta^2} e^{-\zeta\omega t} \sin \left(\omega \sqrt{1-\zeta^2} t + \phi\right)\right\} = \frac{\omega^2}{s(s^2 + 2\zeta\omega s + \omega^2)}, \text{ where } \cos \phi = \zeta$$

$$\int \cot x dx = \ln |\sin x|$$

$$\int \csc x dx = -\ln |\csc x + \cot x|$$

$$\int \sec x dx = -\ln |\sec x + \tan x|$$

Integration by parts:

$$\int u dv = uv - \int v du$$

12.5 Trigonometric identities

Quotient identities:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Reciprocal identities:

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Pythagorean identities:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\csc^2 \theta - \cot^2 \theta = 1$$

Even/odd identities:

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cot(-\theta) = -\cot \theta$$

$$\csc(-\theta) = -\csc \theta$$

$$\sec(-\theta) = \sec \theta$$

Co-function identities:

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan \theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec \theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc \theta$$

$$\frac{\pi}{2} \text{ radians} = 90^\circ$$

Sum/difference identities:

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp$$